

Project Oraculum

Experiments

An issue with outlines-core

outlines-core / src / index.rs

Code

Blame

437 lines (392 loc) · 16.4 KB

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```
... 127         'token_loop: for (token, ids) in vocabulary.tokens().iter() {  
128             if ids.contains(&eos_token_id) {  
129                 continue;  
130             }  
131  
132             let mut next_state = current_state;  
133             for transition_byte in token {  
134                 next_state = dfa.next_state(next_state, *transition_byte);  
135                 if dfa.is_dead_state(next_state) || dfa.is_quit_state(next_state) {  
136                     continue 'token_loop;  
137                 }  
138             }
```

The idea of Ilguidance

- 1) Brzozowski derivatives
- 2) Prefix trees

Brzozowski derivatives

Example #1 - Basic

Start: $(a|b)^*cd$

Derive 'a' $\rightarrow (a|b)^*cd$

Derive 'b' $\rightarrow (a|b)^*cd$

Derive 'c' $\rightarrow d$

Derive 'd' $\rightarrow \epsilon$ Accepting

Example #2 - $(ab)^*$ can be ignored

Start: $(ab)^*c$

Derive 'c' $\rightarrow \epsilon$ Accepting

Brzowski derivatives

Example #3 - However once $(ab)^*$ is started, it needs to be finished

Start: $(ab)^*c$

Derive 'a' $\rightarrow b(ab)^*c$

Derive 'b' $\rightarrow (ab)^*c$

Derive 'a' $\rightarrow b(ab)^*c$

Derive 'b' $\rightarrow (ab)^*c$

Derive 'c' $\rightarrow \varepsilon$ Accepting

Example #4 - Dead end

Start: abc

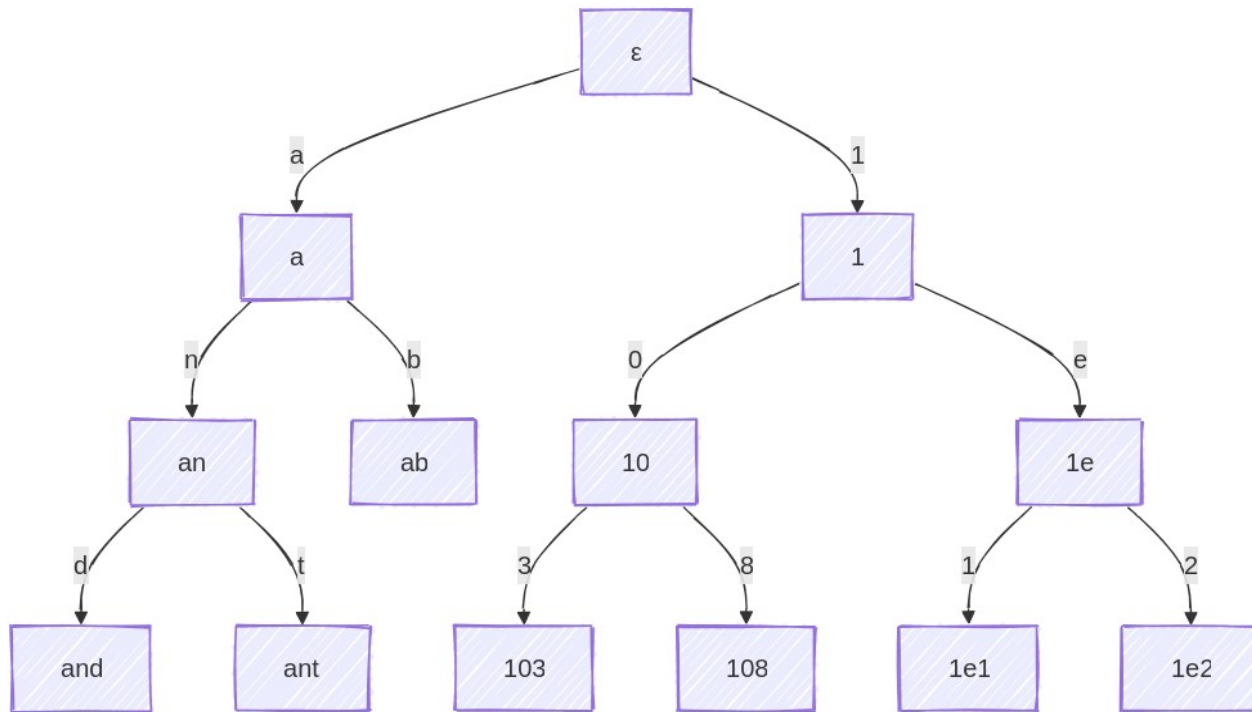
Derive 'a' $\rightarrow bc$

Derive 'b' $\rightarrow c$

Derive 'z' $\rightarrow \emptyset$

Prefix trees

where each node is a token, and each edge is a byte



if “an” is a dead state **than** everything with the “an” prefix is also dead

Significant speed-up

```
Loaded vocabulary in 804.465193ms  
Built outlines vocabulary in 211.950862ms  
Enter regex pattern (press Enter for default):  
Using pattern: (monday|tuesday|wednesday|thursday|friday)+  
Built index in 1.190878411s
```

```
Loaded vocabulary in 786.150486ms  
Built trie in 399.975656ms  
Enter regex pattern (press Enter for default):  
Using pattern: (monday|tuesday|wednesday|thursday|friday)+  
Built index in 4.334894ms
```

Extended prefix trees - Aho-Corasick

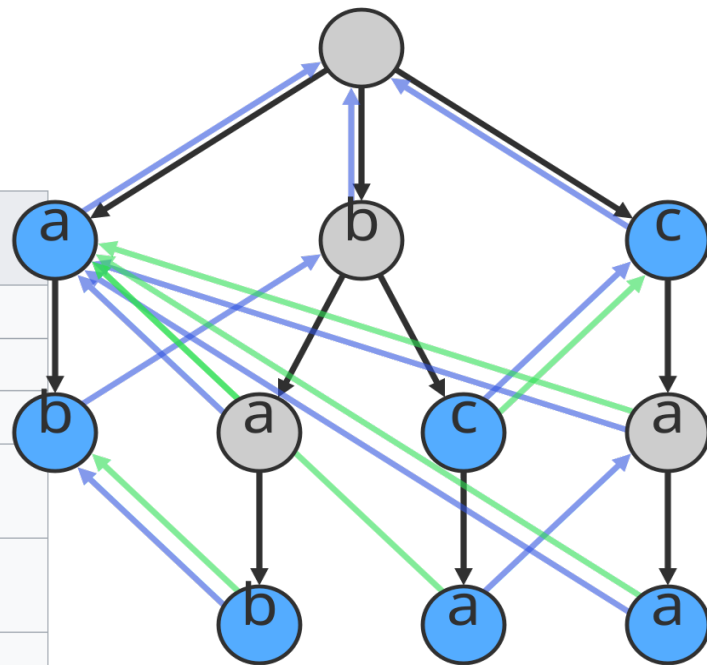
```
Loaded vocabulary in 1.013934923s
Vocabulary size: 255386 tokens
Build time: 2.263387621s
Memory usage: 22.16 MB
Define constant: "It has snowed a lot in Europe"
Lattice construction time: 77.82µs
Lattice memory usage: 1.0703125 KB
```

Input: abccab

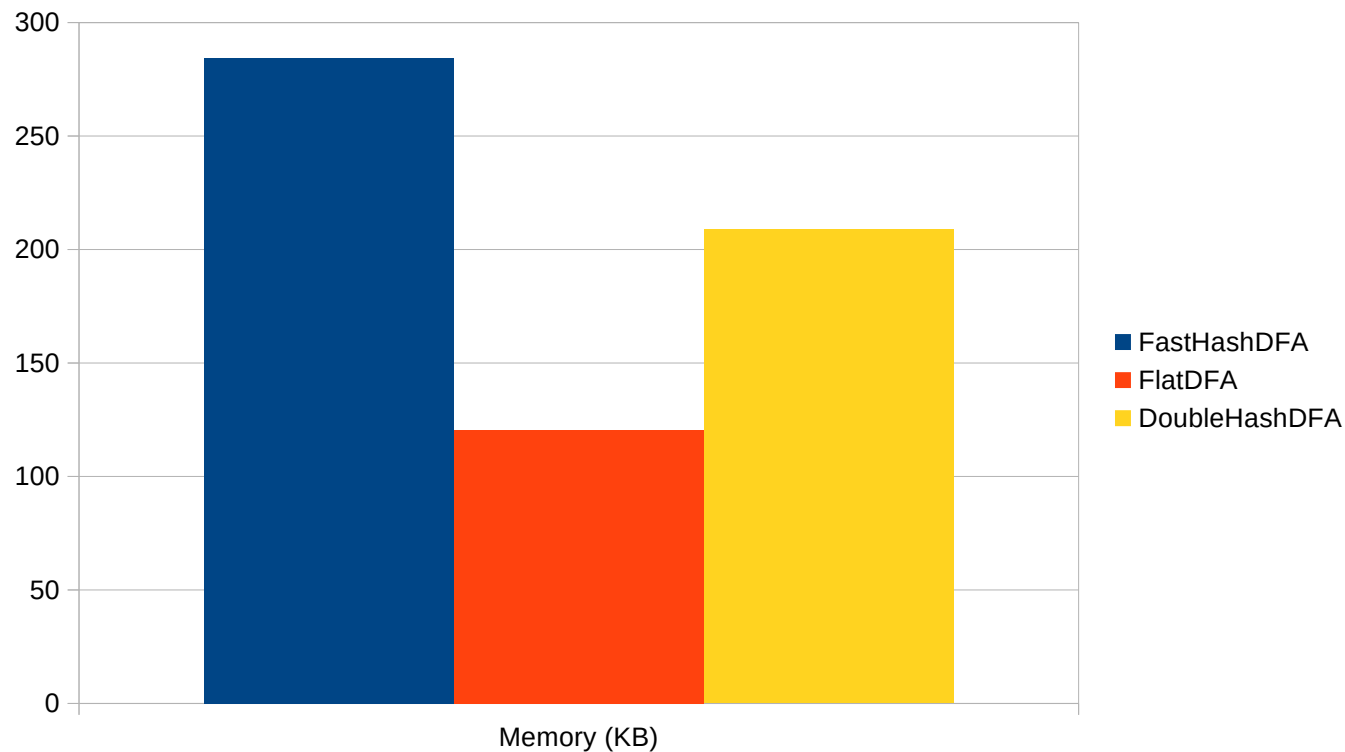
Dictionary

a
ab
bab
bc
bca
c
caa

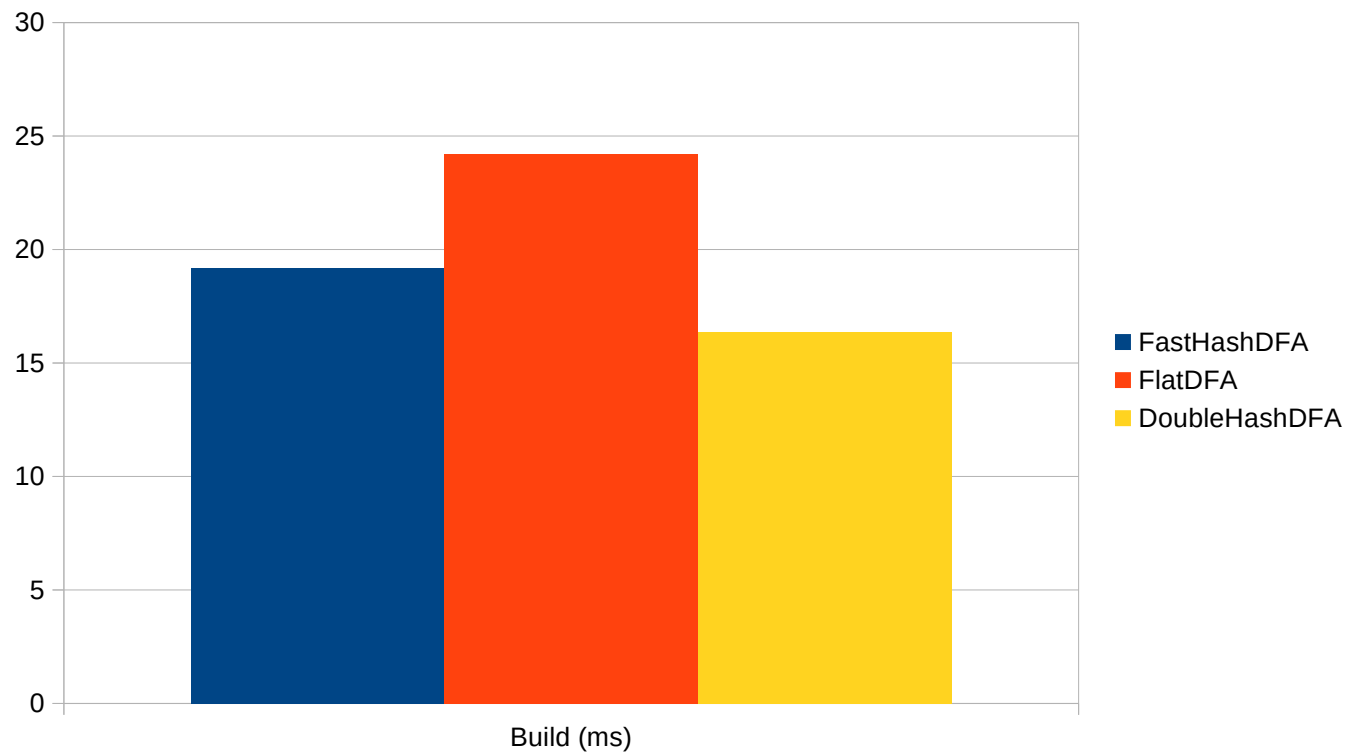
Node	Remaining string	Output: end position	Transition	Output
()	abccab		start at root	
(a)	bccab	a:1	() to child (a)	Current node
(ab)	ccab	ab:2	(a) to child (ab)	Current node
(bc)	cab	bc:3, c:3	(ab) to suffix (b) to child (bc)	Current Node, Dict suffix node
(c)	ab	c:4	(bc) to suffix (c) to suffix () to child (c)	Current node
(ca)	b	a:5	(c) to child (ca)	Dict suffix node
(ab)		ab:6	(ca) to suffix (a) to child (ab)	Current node



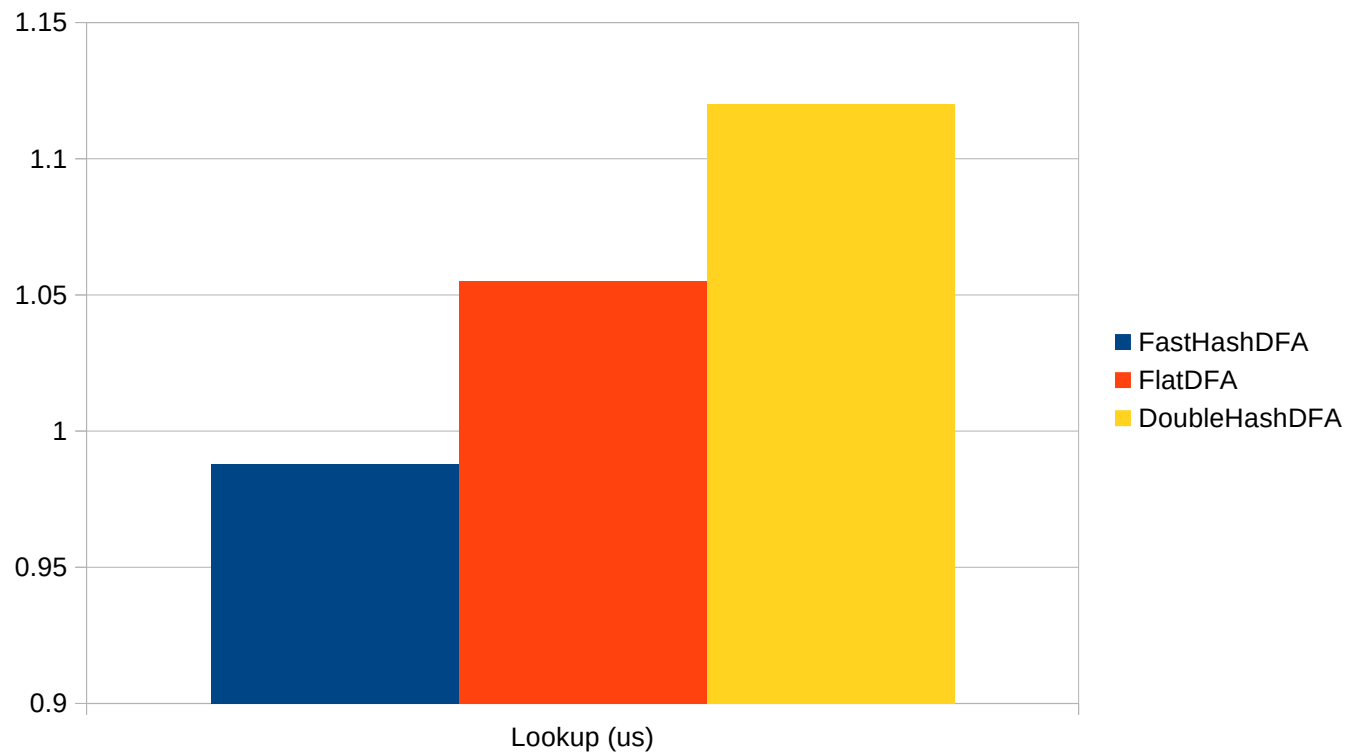
Data structure optimizations



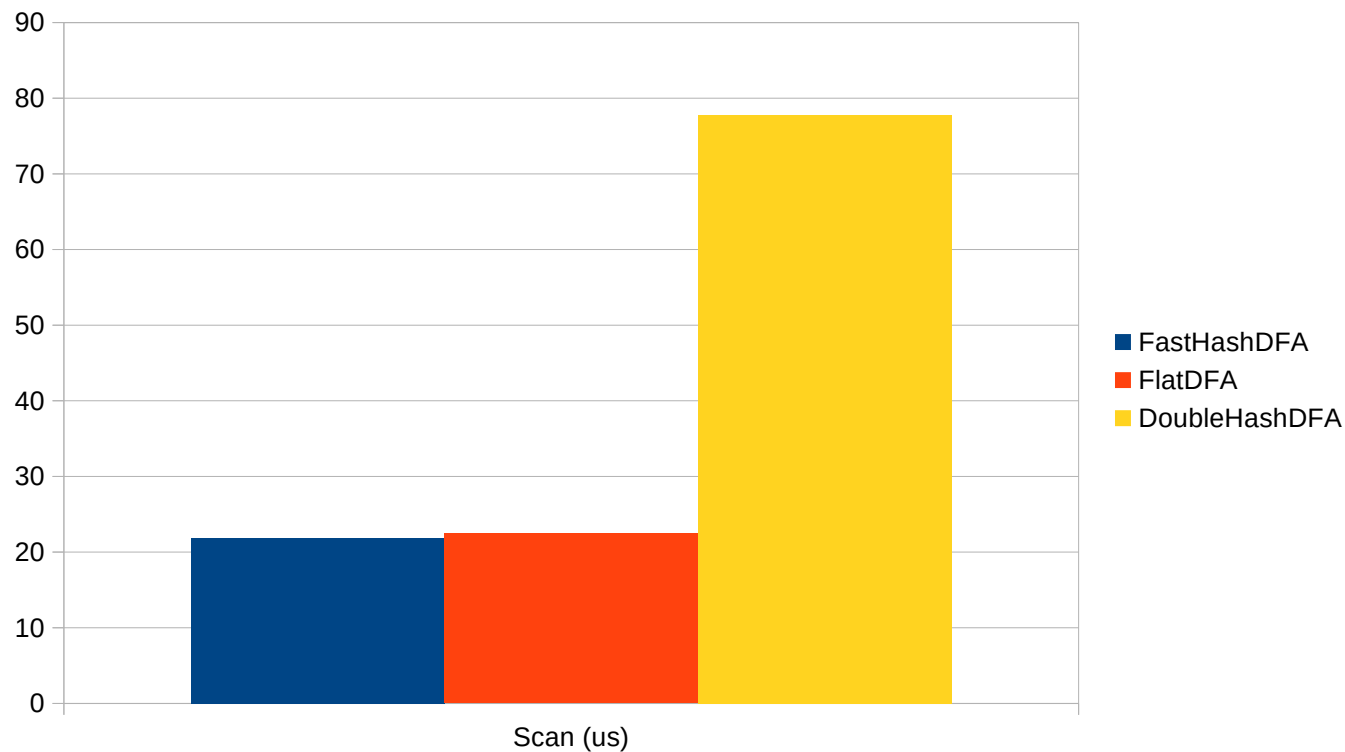
Data structure optimizations



Data structure optimizations



Data structure optimizations



+1 SQL syntactic - semantic enforcer

```
'S', 'SE', 'SEL', 'SELECT'
> 'SEL'
'E', 'EC', 'ECT'
> 'ECT'
',', ',', ', ...
>
'id', 'name', 'email', 'created_at', 'id', 'user_id', 'total', 'status', ...
> 'total'
',', ',', ', ...
>
'id', 'name', 'email', 'created_at', 'id', 'user_id', 'total', 'status'
> 'id'
',', ',', ', ...
>
'id', 'user_id', 'total', 'status'
> 'status'
',', ',', ', ...
>
'F', 'FRO', 'FROM', ...
> 'FROM'
',', ',', ', ...
>
'orders', 'order', 'ord', 'or', ...
> 'orders'
';'
>
'SELECT total, id, status FROM orders;'
```

"total" selected, table "orders" is needed and no table is filtered.

"id" selected, as tables are "orders" needed and "id" is conflicting between "users" and "orders", table "users" is filtered.

Only fields of "orders" are displayed.

Selecting "orders" table is enforced.

```
CREATE TABLE users (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  email TEXT UNIQUE NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE orders (
  id INTEGER PRIMARY KEY,
  user_id INTEGER REFERENCES users(id),
  total DECIMAL(10, 2),
  status TEXT DEFAULT 'pending'
);
```